GILT is a critical host factor for Listeria monocytogenes infection.

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Listeria monocytogenes is a gram-positive, intracellular, food-borne pathogen that can cause severe illness in humans and animals. On infection, it is actively phagocytosed by macrophages; it then escapes from the phagosome, replicates in the cytosol, and subsequently spreads from cell to cell by a non-lytic mechanism driven by actin polymerization. Penetration of the phagosomal membrane is initiated by the secreted haemolysin listeriolysin O (LLO), which is essential for vacuolar escape in vitro and for virulence in animal models of infection. Reduction is required to activate the lytic activity of LLO in vitro, and we show here that reduction by the enzyme gamma-interferon-inducible lysosomal thiol reductase (GILT, also called Ifi30) is responsible for the activation of LLO in vivo. GILT is a soluble thiol reductase expressed constitutively within the lysosomes of antigen-presenting cells, and it accumulates in macrophage phagosomes as they mature into phagolysosomes. The enzyme is delivered by a mannose-6-phosphate receptor-dependent mechanism to the endocytic pathway, where amino- and carboxy-terminal pro-peptides are cleaved to generate a 30-kDa mature enzyme. The active site of GILT contains two cysteine residues in a CXXC motif that catalyses the reduction of disulphide bonds. Mice lacking GILT are deficient in generating major histocompatibility complex class-II-restricted CD4(+) T-cell responses to protein antigens that contain disulphide bonds. Here we show that these mice are resistant to L. monocytogenes infection. Replication of the organism in GILT-negative macrophages, or macrophages expressing an enzymatically inactive GILT mutant, is impaired because of delayed escape from the phagosome. GILT activates LLO within the phagosome by the thiol reductase mechanism shared by members of the thioredoxin family. In addition, purified GILT activates recombinant LLO, facilitating membrane permeabilization and red blood cell lysis. The data show that GILT is a critical host factor that facilitates L. monocytogenes infection.

Interactions of Listeria monocytogenes with mammalian cells during entry and actin-based movement: bacterial factors, cellular ligands and signaling.

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Although <50 kb of its 3.3 megabase genome is known, Listeria monocytogenes has received much attention and an impressive amount of data has contributed in raising this bacterium among the best understood intracellular pathogens. The mechanisms that Listeria uses to enter cells, escape from the phagocytic vacuole and spread from one cell to another using an actin-based motility process have been analysed in detail. Several bacterial proteins contributing to these events have been identified, including the invasion proteins internalin A (InlA) and B (InlB), the secreted pore-forming toxin listeriolysin O (LLO) which promotes the escape from the phagocytic vacuole, and the surface protein ActA which is required for actin polymerization and bacterial movement. While LLO and ActA are critical for the infectious process and are not redundant with other listerial proteins, the precise role of InlA and InlB in vivo remains unclear. How InlA, InlB, LLO or ActA interact with the mammalian cells is beginning to be deciphered. The picture that emerges is that this bacterium uses general strategies also used by other invasive bacteria but has evolved a panel of specific tools and tricks to exploit mammalian cell functions. Their study may lead to a better understanding of important questions in cell biology such as ligand receptor signalling and dynamics of actin polymerization in mammalian cells.

Involvement of reactive oxygen intermediate in the enhanced expression of virulence-associated genes of Listeria monocytogenes inside activated macrophages.

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Listeriolysin O encoded by 1,587 bp hly is the essential virulence factor of Listeria monocytogenes that replicates in the cytosolic space after escaping from phagosome of macrophages. By using murine macrophage-like J774.1 cells with or without activation by IFN-gamma plus LPS, the expression of both hly and its positive regulator prfA was monitored by means of RT-PCR. In activated J774.1 cells, the level of hly expression was enhanced although the multiplication of bacteria was significantly suppressed. The elevated expression of hly inside activated macrophage was abolished by addition of SOD and catalase, suggesting that reactive oxygen intermediates contribute to the upregulation of prfA and hly transcriptions. Moreover, we found that exposure of L. monocytogenes to H2O2 dramatically enhanced the expression of both prfA and hly mRNAs. Spontaneous ONOO- generator, SIN-1, also promoted the transcription to a certain level. These results suggested that oxygen radicals generated in activated macrophages provide a positive signal for up-regulation of virulence genes in L. monocytogenes.

Listeriolysin O-dependent bacterial entry into the cytoplasm is required for calpain activation and interleukin-1 alpha secretion in macrophages infected with Listeria monocytogenes.

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Listeriolysin O (LLO), an hly-encoded cytolysin of Listeria monocytogenes, plays an essential role in the entry of L. monocytogenes into the host cell cytoplasm. L. monocytogenes-infected macrophages produce various proinflammatory cytokines, including interleukin-1 alpha (IL-1 alpha), that contribute to the host immune response. In this study, we have examined IL-1 alpha production in macrophages infected with wild-type L. monocytogenes or a nonescaping mutant strain deficient for LLO (Delta hly). Expression of IL-1 alpha mRNA and accumulation of pro-IL-1 alpha in the cytoplasm were induced by both strains. In contrast, the secretion of the mature form of IL-1 alpha from infected macrophages was observed in infection with wild-type L. monocytogenes but not with the Delta hly mutant. A recovery of the ability to induce IL-1 alpha secretion was shown in a mutant strain complemented with the hly gene. The Toll-like receptor (TLR)/MyD88 signaling pathway was exclusively required for the expression of pro-IL-1 alpha, independently of LLO-mediated cytoplasmic entry of L. monocytogenes. The LLO-dependent secretion of mature IL-1 alpha was abolished by addition of calcium chelators, and only LLO-producing L. monocytogenes strains were able to induce elevation of the intracellular calcium level in infected macrophages. A calcium-dependent protease, calpain, was implicated in the maturation and secretion of IL-1 alpha induced by LLO-producing L. monocytogenes strains based on the effect of calpain inhibitor. Functional activation of calpain was detected in macrophages infected with LLO-producing L. monocytogenes strains but not with a mutant strain lacking LLO. These results clearly indicated that LLO-mediated cytoplasmic entry of bacteria could induce the activation of intracellular calcium signaling, which is essential for maturation and secretion of IL-1 alpha in macrophages during L. monocytogenes infection through activation of a calcium-dependent calpain protease. In addition, recombinant LLO, when added to macrophages infected with the Delta hly strain, could induce calcium influx and IL-1 alpha secretion at doses exhibiting cytolytic activity, suggesting that LLO produced by intracellular L. monocytogenes may be implicated in induction of calcium influx through pore formation.

Regulated translation of listeriolysin O controls virulence of Listeria monocytogenes.

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Listeria monocytogenes is an intracytosolic bacterial pathogen that escapes from the phagosome using a secreted cytolysin, listeriolysin O (LLO). In the host cytosol, LLO activity is minimized to prevent pore formation in the host plasma membrane; premature lysis of the infected host cell exposes the bacteria to extracellular immune defences of the host and is detrimental to infection. Here we identified nucleotide substitutions in the coding sequence of the LLO gene (hly) that did not alter the protein sequence, yet caused over-production of LLO, cytotoxicity and loss of virulence. These phenotypes were independent of the promoter and, under conditions in which the mutants produced more LLO protein than wild type, levels of hly mRNA were similar. Finally, negative regulation of LLO was maintained even when bacteria were engineered to produce elevated levels of the wild-type hly transcript. Together, our data demonstrate that translational regulation of LLO is critical for L. monocytogenes pathogenesis.

Listeriolysin O as a reporter to identify constitutive and in vivo-inducible promoters in the pathogen Listeria monocytogenes.

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Listeria monocytogenes is a facultative intracellular gram-positive bacterium capable of growing in the cytoplasm of infected host cells. Bacterial escape from the phagosomal vacuole of infected cells is mainly mediated by the pore-forming hemolysin listeriolysin O (LLO) encoded by hly. LLO-negative mutants of L. monocytogenes are avirulent in the mouse model. We have developed a genetic system with hly as a reporter gene allowing the identification of both constitutive and in vivo-inducible promoters of this pathogen. Genomic libraries were created by randomly inserting L. monocytogenes chromosomal fragments upstream of the promoterless hly gene cloned into gram-positive and gram-negative shuttle vectors and expressed in an LLO-negative mutant strain. With this hly-based promoter trap system, combined with access to the L. monocytogenes genome database, we identified 20 in vitro-transcribed genes, including genes encoding (i) p60, a previously known virulence gene, (ii) a putative new hemolysin, and (iii) two proteins of the general protein secretion pathway. By using the hly-based system as an in vivo expression technology tool, nine in vivo-induced loci of L. monocytogenes were identified, including genes encoding (i) the previously known in vivo-inducible phosphatidylinositol phospholipase C and (ii) a putative N-acetylglucosamine epimerase, possibly involved in teichoic acid biosynthesis. The use of hly as a reporter is a simple and powerful alternative to classical methods for transcriptional analysis to monitor promoter activity in L. monocytogenes.

The broad-range phospholipase C and a metalloprotease mediate listeriolysin O-independent escape of Listeria monocytogenes from a primary vacuole in human epithelial cells.

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Intracellular growth of Listeria monocytogenes begins after lysis of the primary vacuole formed upon bacterial entry into a host cell. Listeriolysin O (LLO), a pore-forming hemolysin encoded by hly, is essential for vacuolar lysis in most cell types. However, in human epithelial cells, LLO- mutants are capable of growth, suggesting that gene products other than LLO are capable of mediating escape from a vacuole. In this study, we investigated the role of other bacterial gene products in lysis of the primary vacuole in the human epithelial cell line Henle 407. Double internal in-frame deletion mutants were constructed by introducing a mutated hly allele into strains harboring deletions in either of the phospholipase C (PLC)-encoding genes or a metalloprotease-encoding gene. Bacterial escape from the primary vacuole, intracellular growth, and cell-to-cell spread were evaluated in Henle 407 cells. The results indicated that, in the absence of LLO, the broad-range PLC and the metalloprotease were both required for lysis of the primary vacuole in Henle 407 cells. Although phosphatidylinositol-specific PLC was not required, the efficiency of escape was reduced in an LLO phosphatidylinositol-specific PLC double mutant. These observations suggest that the relative importance of LLO, the phospholipases, and the metalloprotease may vary in different cell types or in cells from different species. In addition, these studies provide insight into the mechanisms of action of virulence determinants involved in the lysis of vacuolar membranes.

Biological effects of listeriolysin O: implications for vaccination.

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Listeriolysin O (LLO) is a thiol-activated cholesterol-dependent pore-forming toxin and the major virulence factor of Listeria monocytogenes (LM). Extensive research in recent years has revealed that LLO exerts a wide array of biological activities, during the infection by LM or by itself as recombinant antigen. The spectrum of biological activities induced by LLO includes cytotoxicity, apoptosis induction, endoplasmic reticulum stress response, modulation of gene expression, intracellular calcium oscillations, and proinflammatory activity. In addition, LLO is a highly immunogenic toxin and the major target for innate and adaptive immune responses in different animal models and humans. Recently, the crystal structure of LLO has been published in detail. Here, we review the structure-function relationship for this fascinating microbial molecule, highlighting the potential uses of LLO in the fields of biomedicine and biotechnology, particularly in vaccination.

Listeria infection modulates mitochondrial dynamics.

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Mitochondria are highly dynamic organelles that are central to several cellular processes, the most prominent being energy production. Several reports have shown that pathogens target mitochondria in various ways to interfere with apoptosis, but to our knowledge only one study has specifically addressed the effects of infection on mitochondrial dynamics. We focused on this aspect during infection with the intracellular pathogen L. monocytogenes and could recently show that this bacterium profoundly alters mitochondrial dynamics, causing transient fragmentation of the mitochondrial network. This mitochondrial fragmentation occurs early during infection and is specific to pathogenic L. monocytogenes, as it is not observed with other intracellular pathogens. The relevance of mitochondrial dynamics for L. monocytogenes infection is highlighted by the finding that siRNA-mediated inhibition of mitochondrial fusion or fission decreases or increases the efficiency of L. monocytogenes infection, respectively. The main bacterial factor responsible for mitochondrial network disruption was identified as the secreted pore-forming toxin listeriolysin O, which also appeared to impair mitochondrial function. Our work suggests that in order to establish an efficient infection, L. monocytogenes interferes with cellular physiology at early timepoints by transient disruption of mitochondrial dynamics and function.

Cytolysin-dependent escape of the bacterium from the phagosome is required but not sufficient for induction of the Th1 immune response against Listeria monocytogenes infection: distinct role of Listeriolysin O determined by cytolysin gene replacement.

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Listeria monocytogenes evades the antimicrobial mechanisms of macrophages by escaping from the phagosome into the cytosolic space via a unique cytolysin that targets the phagosomal membrane, listeriolysin O (LLO), encoded by hly. Gamma interferon (IFN-gamma), which is known to play a pivotal role in the induction of Th1-dependent protective immunity in mice, appears to be produced, depending on the bacterial virulence factor. To determine whether the LLO molecule (the major virulence factor of L. monocytogenes) is indispensable or the escape of bacteria from the phagosome is sufficient to induce IFN-gamma production, we first constructed an hly-deleted mutant of L. monocytogenes and then established isogenic L. monocytogenes mutants expressing LLO or ivanolysin O (ILO), encoded by ilo from Listeria ivanovii. LLO-expressing L. monocytogenes was highly capable of inducing IFN-gamma production and Listeria-specific protective immunity, while the hly-deleted mutant was not. In contrast, the level of IFN-gamma induced by ILO-expressing L. monocytogenes was significantly lower both in vitro and in vivo, despite the ability of this strain to escape the phagosome and the intracellular multiplication at a level equivalent to that of LLO-expressing L. monocytogenes. Only a negligible level of protective immunity was induced in mice against challenge with LLO- and ILO-expressing L. monocytogenes. These results clearly show that escape of the bacterium from the phagosome is a prerequisite but is not sufficient for the IFN-gamma-dependent Th1 response against L. monocytogenes, and some distinct molecular nature of LLO is indispensable for the final induction of IFN-gamma that is essentially required to generate a Th1-dependent immune response.